

SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 04: Marginal and Conditional Probability Distributions

LECTURE W4L1: Bivariate Joint Probability Distributions

BY

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CONCEPTS COVERED

Bivariate Joint Probability Distributions: Discrete and Continuous

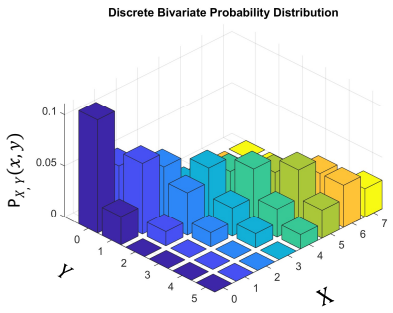
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Bivariate Probability Distribution

- In many real-world problems, two or more variables need to be analyzed together because they may occur simultaneously or influence each other.
- A bivariate probability distribution describes the probability behavior of two random variables considered jointly.
- These distributions are essential for analyzing interconnected real-world events. For example, to study the relationship between: temperature and relative humidity; air pollution levels and respiratory illness cases; GPU memory errors and model retraining failures etc.
- Bivariate distributions are classified into discrete and continuous types, depending on whether the variables take countable values or fall within specified continuous range.

Discrete Bivariate Random Variable

- Distribution functions involving more than one random variables are called multivariate probability distribution:
  - Two variables → Bivariate distributions
  - Three variables → Trivariate distributions.
- In this lecture we learn bivariate probability distributions for both discrete and continuous random variables.
- If X and Y are two discrete random variables, then the pair: (X, Y) is used in a bivariate **Joint Probability Mass Function**.
- Both X and Y can take only specific values, such as 0, 1, 2, 3, as shown in the figure.

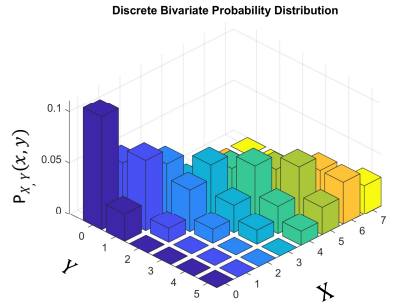


Joint Probability Mass Function

- Let us consider X and Y are two discrete random variables and let $p_{X,Y}(x,y)$ be their joint probability mass function (PMF).
- For a valid joint PMF of two discrete random variables X and Y, following conditions are to be fulfilled:

$$P_{X,Y}(x,y) \geq 0 \quad \text{for all } x \text{ and } y, \text{ and}$$

$$\sum_{\text{all } x} \sum_{\text{all } y} P_{X,Y}(x,y) = 1$$



- A typical plot of a discrete bivariate PMF is shown in the figure, where each vertical bar represents the probability mass concentrated at a specific ordered pair (x,y).
- The height of each bar shows the joint probability $p_{X,Y}(x,y)$, showing how likely a particular combination of X and Y is to occur.

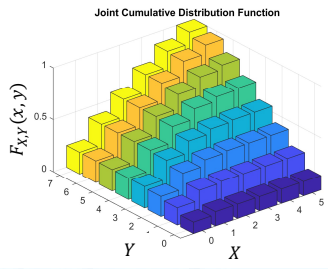
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Joint Cumulative Distribution Function

- If $F_{X,Y}(x,y)$ be the corresponding cumulative probability distribution function, then,

$$F_{X,Y}(x,y) = P(X \leq x_i, Y \leq y_j)$$

$$= \sum\sum P_{X,Y}(x_i,y_j) \text{ for all } (x_i,y_j) \text{ such that } x_i < x \text{ and } y_j < y$$



- A typical plot of a joint cumulative distribution function, (joint CDF), is shown in the figure, where each vertical bar represents the cumulative probability up to a specific ordered pair (x,y).
- The joint CDF plot shows the accumulated probability up to that point.
- Therefore, the bar heights generally increase as we move toward larger values of X and Y, and the maximum cumulative probability approaches 1.

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Example – 1

In a cloud-based Machine Learning environment, two discrete random variables are monitored during a weekly model deployment cycle: X = number of GPU out-of-memory (OOM) errors in a week, Y = number of automated model retraining failures in the same week. The joint PMF of X and Y is given by:

$$P_{X,Y}(x,y) = k(2x + y + 1) \text{ where: } x = 0,1,2,3 \text{ and } y = 0,1,2$$

a) Find the value of k, so that $P_{X,Y}(x,y)$ is a valid joint PMF.

b) Find the probability that a week has exactly 2 GPU OOM errors and exactly 1 model retraining failure.

c) Find the probability that a week has more model retraining failures than GPU OOM errors.

Solution

a) Finding value of k

For a valid joint PMF, $\sum_{x=0}^{3}\sum_{y=0}^{2} P_{X,Y}(x,y) = 1$

$$\Rightarrow \sum_{x=0}^{3}\sum_{y=0}^{2} k(2x + y + 1) = 1$$

Now, preparing the table for all possible combinations of x and y:

$X\backslash Y$	0	1	2
0	k	$2k$	$3k$
1	$3k$	$4k$	$5k$
2	$5k$	$6k$	$7k$
3	$7k$	$8k$	$9k$

$X\backslash Y$	0	1	2
0	1/60	2/60	3/60
1	3/60	4/60	5/60
2	5/60	6/60	7/60
3	7/60	8/60	9/60

Sum of all values:

$$k + 2k + 3k + 3k + 4k + 5k + 5k + 6k + 7k + 7k + 8k + 9k = 60k$$

Therefore, $60k = 1 \Rightarrow k = \frac{1}{60}$

Solution

b) Probability that a week has exactly 2 GPU OOM errors and exactly 1 model retraining failure

$$P(X = 2, Y = 1) = k(2(2) + 1 + 1) = \frac{1}{60}(6) = \frac{1}{10} = 0.1$$

$X \backslash Y$	0	1	2
0	1/60	2/60	3/60
1	3/60	4/60	5/60
2	5/60	6/60	7/60
3	7/60	8/60	9/60

c) Probability that a week has more model retraining failures than GPU OOM errors

$P(Y > X)$

Possible pairs satisfying $Y > X$ are: $(0,1), (0,2), (1,2)$

$$P(Y > X) = P(0,1) + P(0,2) + P(1,2) = \frac{1}{60}(2 + 3 + 5) = \frac{10}{60}$$

$$= 0.167$$

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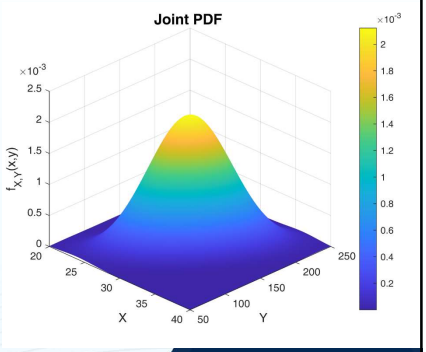
Continuous Bivariate Random Variable

- If X and Y are two continuous random variables, then the pair (X, Y) is called a continuous bivariate random variable.
- Both X and Y can take any value within a specified interval.
- For example,

  X = Amount of nitrogen-based fertilizer applied per hectare (in kg)

  Y = Total crop yield per hectare (in tonnes)

  So, the pair: (X, Y) = (120, 8.2) means applying 120 kg of fertilizer to a hectare resulted in an 8.2 tonnes crop yield.



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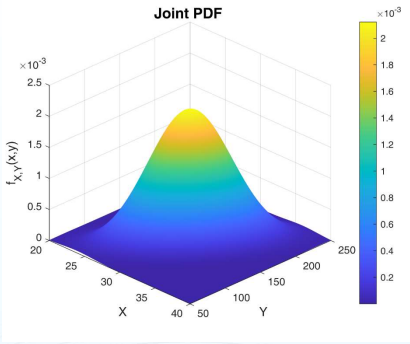
Joint Probability Density Function

- Let us consider X and Y are two continuous random variables and let $f_{X,Y}(x,y)$ be their joint probability density function (PDF).
- For a valid joint PDF of two continuous random variables X and Y, following conditions are to be fulfilled:

$$f_{X,Y}(x,y) \geq 0 \quad \text{for all } x \text{ and } y, \text{ and}$$

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f_{X,Y}(x,y)\, dx\, dy = 1$$

- A typical plot of a continuous bivariate PDF is shown in the figure.
- In continuous bivariate distributions, probability is obtained by calculating the volume under the PDF surface over a specified region.



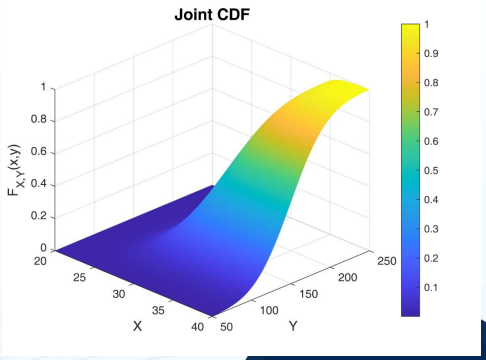
Joint Cumulative Distribution Function

- If $F_{X,Y}(x,y)$ be the corresponding cumulative probability distribution function, then,

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$
$$= \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(t,s)\, ds\, dt$$

- A typical plot of a joint cumulative distribution function, CDF, is shown in the figure.
- The PDF $f_{X,Y}(x,y)$ and the CDF $F_{X,Y}(x,y)$ are related as follows:

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$$



Example – 2

In an automated Machine Learning pipeline, the daily time required to update a predictive model fluctuates based on the volume of incoming data. Two continuous random variables are observed:

X = time spent on data preprocessing (in min)

Y = time spent on model training (in min)

Suppose the joint PDF of X and Y is given by,

$$f_{X,Y}(x,y) = \begin{cases} Cxy & \text{for } 20 < x < 30, 30 < y < 50 \\ 0 & elsewhere \end{cases}$$

a) Find the value of C so that $f_{X,Y}(x,y)$ is a valid joint PDF.

b) Find the probability that the data preprocessing takes between 20 and 25 minutes, and the model training takes between 30 and 40 minutes.

Solution

a) For a valid joint PDF: $\int_{20}^{30}\int_{30}^{50} C\,xy\,dy\,dx = 1$

Now, $\int_{20}^{30}\int_{30}^{50} C\,xy\,dy\,dx = \int_{20}^{30} Cx.dx \left[\frac{y^2}{2}\right]_{30}^{50} = \int_{20}^{30} Cx.dx \left(\frac{50^2-3\ ^2}{2}\right)$

$= \int_{20}^{30} Cx.dx(\frac{2500-900}{2}) = \int_{20}^{30} 800Cx.dx = 800\,C\left[\frac{x^2}{2}\right]_{20}^{30} = 800C\left(\frac{30^2-2\ ^2}{2}\right)$

$= 800C\left(\frac{900-400}{2}\right) = 800C(250) = 200000C$

$\Rightarrow 200000C = 1 \Rightarrow C = \frac{1}{2\times10^5}$

So, the PDF is – $f_{X,Y}(x,y) = \begin{cases} \frac{xy}{2\times10^5} & \text{for } 20 < x < 30, 30 < y < 50 \\ 0, & elsewhere \end{cases}$

Solution

b) Probability that the data preprocessing takes between 20 and 25 minutes, and the GPU model training takes between 30 and 40 minutes

$$P(20 < X < 25, 30 < Y < 40) = \int_{20}^{25} \int_{30}^{40} \frac{xy}{2\times10^5} dy\, dx$$

First integrating with respect to y:

$$\int_{30}^{40} \frac{xy}{2\times10^5} dy = \frac{x}{2\times10^5}\left[\frac{y^2}{2}\right]_{30}^{40} = \frac{x}{2\times10^5}\left(\frac{40^2-30^2}{2}\right)$$

$$= \frac{x}{2\times10^5}\left(\frac{1600-900}{2}\right) = \frac{350x}{2\times10^5}$$

Solution

Now integrating with respect to x:

$$\int_{20}^{25} \frac{350x}{2\times10^5} dx = \frac{350}{2\times10^5}\left[\frac{x^2}{2}\right]_{20}^{25} = \frac{350}{2\times10^5}\left(\frac{25^2-20^2}{2}\right)$$

$$= \frac{350}{2\times10^5}\left(\frac{625-4}{2}\right) = \frac{350}{2\times10^5} \times (112.5) = 0.197$$

Therefore, $P(20 < X < 25, 30 < Y < 40) = 0.197$

The probability that the data preprocessing takes between 20 and 25 minutes, and the model training takes between 30 and 40 minutes is 0.197.

Joint Cumulative Distribution Function

- Joint CDF represents the accumulated probability up to a particular combination of X and Y.
- Some of the properties of continuous bivariate cumulative distribution are -
  i. $F_{X,Y}(x, \infty)$ is the cumulative marginal probability function of X.
  ii. $F_{X,Y}(\infty, y)$ is the cumulative marginal probability function of Y.
  iii. $F_{X,Y}(\infty, \infty) = 1$
  iv. $F_{X,Y}(-\infty, y) = F_{X,Y}(x, -\infty) = 0$

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MODULE 04 : Marginal and Conditional Probability Distributions

LECTURE W4L2 : Marginal Probability Distributions for Discrete RV

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CONCEPTS COVERED

Marginal Probability Distributions: Discrete Random Variables

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Marginal Probability Distribution

- Marginal probability distribution is an important concept in multivariate probability distributions.
- These distributions are commonly applied in prediction, modelling, and simulation studies using statistical and machine learning approaches.
- Many real-world systems often involve multiple variables. However, analysis of a single one is required in many cases.
- Marginal distribution is derived from joint density (pmf/pdf) by marginalizing out all other variables.
- This concept will be explained in the context of bivariate distributions involving two random variables (either discrete or continuous).
- The same concept can later be extended to multivariate distributions with more than two random variables.

Marginal Probability Distribution

Discrete Random Variable:

- Let us consider X and Y be two discrete random variables with their joint probability mass function (pmf) as $p_{X,Y}(x, y)$.
- The joint distribution with $p_{X,Y}(x_i, y_j) = P(X = x_i, Y = y_j)$ for $i = 1,2, \dots, m$ and $j = 1,2, \dots, n$ appears as a $m \times n$ 2-dimensional table of probability values corresponding to a pair of X and Y values.
- Marginal distribution represents the probability distribution of one of the two discrete random variables, either X or Y irrespective of the distribution of the other variable.

Discrete Random Variable:

- Thus, the marginal distribution of X is obtained by summing up the probability values for all possible values of Y, which is also known as marginalized out.
- Mathematically, it is obtained from $p_{X,Y}(x_i, y_j) = P(X = x_i, Y = y_j)$ for $i = 1,2,\dots,m$ and $j = 1,2,\dots,n$ as follows:

$$p_X(x_i) = \sum_{k=1}^{n} p_{X,Y}(x_i, y_k) \text{ for } i = 1,2,\dots,m$$

- Similarly, the marginal distribution of Y is obtained as:

$$p_Y(y_j) = \sum_{k=1}^{m} p_{X,Y}(x_k, y_j) \text{ for j} = 1,2,\dots,n$$

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Discrete joint distribution between X and Y along with the marginal distributions of X (last column) and Y (last row)

Random Variables		Y				Marginal distribution of $X(p_x(x_i))$
		y_1	y_2	...	y_n	
X	x_1	$p_{X,Y}(x_1, y_1)$	$p_{X,Y}(x_1, y_2)$	...	$p_{X,Y}(x_1, y_n)$	$p_x(x_1) = \sum_{\text{all } j} p_{X,Y}(x_1, y_j)$
	x_2	$p_{X,Y}(x_2, y_1)$	$p_{X,Y}(x_2, y_2)$	...	$p_{X,Y}(x_2, y_n)$	$p_x(x_2) = \sum_{\text{all } j} p_{X,Y}(x_2, y_j)$
	⋮	⋮	⋮	⋱	⋮	⋮
	x_m	$p_{X,Y}(x_m, y_1)$	$p_{X,Y}(x_m, y_2)$	...	$p_{X,Y}(x_m, y_n)$	$p_x(x_m) = \sum_{\text{all } j} p_{X,Y}(x_m, y_j)$
Marginal distribution of Y $(p_Y(y_j))$		$p_Y(y_1) = \sum_{\text{all } i} p_{X,Y}(x_i, y_1)$	$p_Y(y_2) = \sum_{\text{all } i} p_{X,Y}(x_i, y_2)$	...	$p_Y(y_n) = \sum_{\text{all } i} p_{X,Y}(x_i, y_n)$	

A Typical Example of Joint PMF

	$Y = 0$	$Y = 1$	$Y = 2$	Marginal $p_X(x)$
$X = 0$	0.10	0.15	0.05	0.30
$X = 1$	0.20	0.25	0.10	0.55
$X = 2$	0.05	0.05	0.05	0.15
Marginal $p_Y(y)$	0.35	0.45	0.20	$\sum = 1$

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Cumulative Marginal Probability Distribution

Discrete Random Variable:

- The cumulative marginal distribution of X is

$$F_X(x) = \sum_{x_i \le x} p_X(x_i) = \sum_{x_i \le x \ all \ y_j} p_{X,Y}(x_i, y_j)$$

- Similarly, the cumulative marginal distribution of Y is

$$F_Y(y) = \sum_{y_j \le y} p_Y(y_j) = \sum_{y_j \le y \ all \ x_i} p_{X,Y}(x_i, y_j)$$

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Example – 1

To evaluate the performance of an ensemble classification system predicting seasonal crop yields, the output is categorized into four discrete classes: 1, 2, 3, and 4. Let the random variables X and Y represent the predicted yield classes generated by Model A and Model B, respectively, for a given validation set. The joint pmf of the predictions X and Y is given in the following probability matrix. Find the marginal pmfs and cumulative marginal distributions of the predictions to assess the individual prediction biases of Model A and Model B.

Random Variable		Y			
		1	2	3	4
X	1	0.10	0.05	0.00	0.00
	2	0.05	0.20	0.05	0.00
	3	0.00	0.10	0.25	0.05
	4	0.00	0.00	0.05	0.10

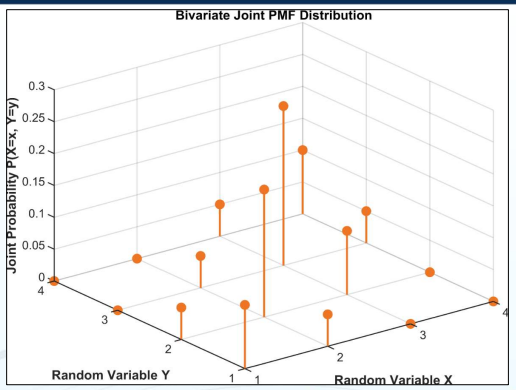
Solution

A joint pmf is valid if all individual probabilities are non-negative and their total sum is equal to 1.

$$\sum\sum P_{X,Y}(x,y)$$

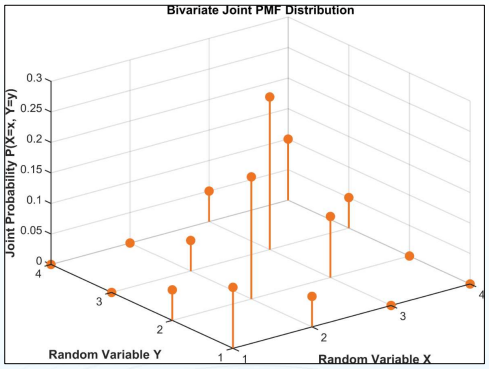
$$= 0.1+0.05+0.05+0.2+0.05$$
$$+0.1+0.25+0.05+0.05+0.1$$
$$= 1$$

Thus, the given joint pmf is valid.



Solution

Marginal distribution of X:



- The marginal probability of X is calculated by summing probabilities over all possible values of Y.

$$p_X(x_i) = \sum_{k=1}^{n} p_{X,Y}(x_i, y_k) \text{ for } i = 1,2,\dots,m$$

- Thus, Marginal pmf of crop yield in Zone A,

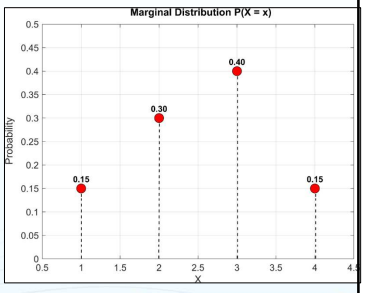
$$p_X(x_i) = \sum_{k=1}^{4} p_{X,Y}(x_i, y_k) \text{ for } i = 1,2,3 \text{ and } 4$$

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Solution

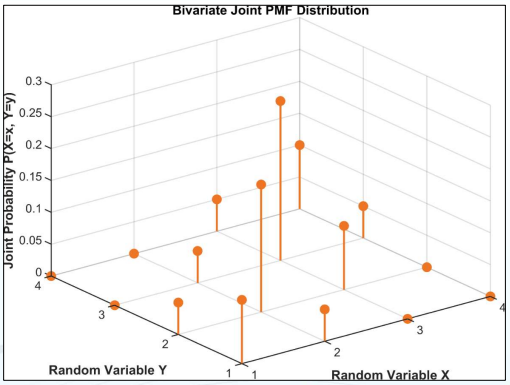
Marginal distribution of X:

Random Variable		Y				Marginal distribution of X	
		1	2	3	4	$\sum_{k=1}^{4} p_{X,Y}(x, y_k)$	
X	1	0.10	0.05	0.00	0.00	$0.10 + 0.05 + 0 + 0$	= 0.15
	2	0.05	0.20	0.05	0.00	$0.05 + 0.20 + 0.05 + 0$	= 0.30
	3	0.00	0.10	0.25	0.05	$0 + 0.1 + 0.25 + 0.05$	= 0.40
	4	0.00	0.00	0.05	0.10	$0 + 0 + 0.05 + 0.1$	= 0.15



Solution

Marginal distribution of Y:



- Similarly, the marginal distribution of Y is obtained by summing over all values of X.

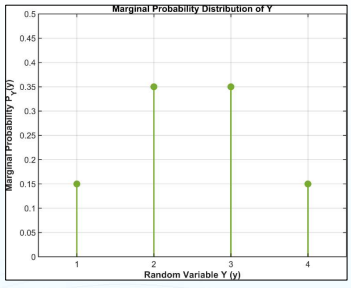
$$p_Y(y_j) = \sum_{k=1}^{m} p_{X,Y}(x_k, y_j) \text{ for } j = 1,2,\dots,n$$

- Thus, Marginal pmf of crop yield in Zone B,

$$p_Y(y_j) = \sum_{k=1}^{4} p_{X,Y}(x_k, y_j) \text{ for } j = 1,2,3,\text{and } 4$$

Solution

Random Variable		Y			
		1	2	3	4
X	1	0.10	0.05	0.00	0.00
	2	0.05	0.20	0.05	0.00
	3	0.00	0.10	0.25	0.05
	4	0.00	0.00	0.05	0.10
	$p_Y(y_j) = \sum_{k=1}^{4} p_{X,Y}(x_k, y_j)$	=0.15	=0.35	=0.35	=0.15



Solution

The cumulative marginal distribution of X is:

$$F_X(x) = \sum_{x_i \le x} p_X(x_i) = \sum_{x_i \le x \text{ all } y_j} p_{X,Y}(x_i, y_j)$$

$$F_X(1) = p_X(1) = 0.15$$

$$F_X(2) = p_X(1) + p_X(2) = 0.15 + 0.30 = 0.45$$

$$F_X(3) = p_X(1) + p_X(2) + p_X(3) = 0.45 + 0.40 = 0.85$$

$$F_X(4) = p_X(1) + p_X(2) + p_X(3) + p_X(4)$$

$$= 0.85 + 0.15 = 1.00$$

Solution

Similarly, the cumulative marginal of Y is calculated as:

$$F_Y(y) = \sum_{y_j \le y} p_Y(y_j) = \sum_{y_j \le y \text{ all } x_i} p_{X,Y}(x_i, y_j)$$

$$F_Y(1) = 0.15$$

$$F_Y(2) = 0.15 + 0.35 = 0.50$$

$$F_Y(3) = 0.50 + 0.35 = 0.85$$

$$F_Y(4) = 0.85 + 0.15 = 1.00$$


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Statistical and Machine Learning Methods: Basics and Applications

MODULE 04 : Marginal and Conditional Probability Distributions

LECTURE W4L3 : Marginal Probability Distributions for Continuous RV

BY

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CONCEPTS COVERED

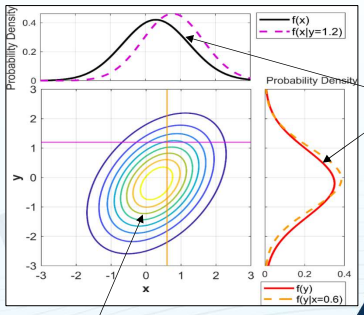
Marginal Probability Distributions : Continuous Random Variables

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Marginal Probability Distribution

Continuous Random Variable:

- Let us consider X and Y to be two continuous random variables with their joint pdf as, $f_{X,Y}(x,y)$.
- Following the similar concept, the marginal distribution of X can be obtained by marginalizing Y out.
- Mathematically, Y is integrated out to get the marginal distribution of X from the joint pdf $f_{X,Y}(x,y)$.



Marginal distributions

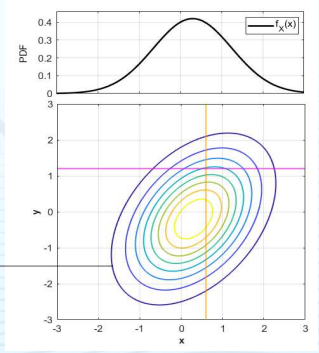
Joint PDF

Marginal Probability Distribution

Continuous Random Variable:

- Thus, the marginal distribution of X is obtained as

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy$$

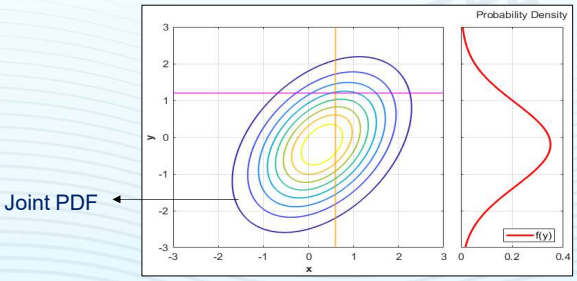


Joint PDF

Continuous Random Variable:

- Similarly, the marginal distribution of Y is obtained by integrating over all values of X.

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dx$$



Cumulative Marginal Probability Distribution

Continuous Random Variable:

- The corresponding cumulative marginal distributions of X and Y are:

$$F_X(x) = \int_{-\infty}^{x} f_X(x)dx = \int_{-\infty}^{x} \left[\int_{-\infty}^{\infty} f_{X,Y}(x,y)dy\right]dx$$

$$F_Y(y) = \int_{-\infty}^{y} f_y(y)dy = \int_{-\infty}^{y} \left[\int_{-\infty}^{\infty} f_{X,Y}(x,y)dx\right]dy$$

Example – 1

During the evaluation of a Deep Neural Network, researchers analyze the relationship between layer activation and prediction confidence. Let X represent the proportion of active neurons in a critical hidden layer (normalized between 0 and 1, where 1 means all neurons are firing). Let Y represent the confidence score for the network's final prediction (ranging from 0 to 1). The joint probability density function (PDF) capturing their relationship is given by:

$$f\ (x,y) = 6x^2y, \qquad 0 \le x \le 1, 0 \le y \le 1$$

Find the marginal distribution functions and the marginal cumulative distribution functions for both the active neuron proportion (X) and the confidence score (Y).

Solution

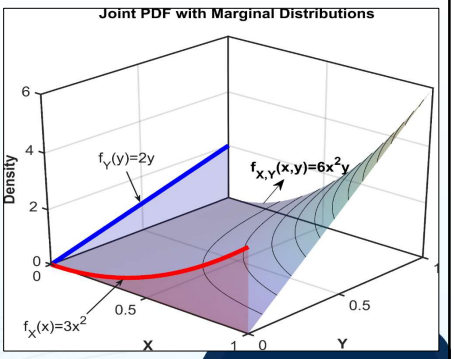
Given joint pdf,

$$f\ (x,y) = 6x^2y, \qquad 0 \le x \le 1, 0 \le y \le 1$$

Marginal distribution function of X,

$$f_X(x) = \int_0^1 f_{X,Y}\ (x,y)\, dy$$

$$= \int_0^1 6x^2y\, dy = \frac{6x^2}{2}[y^2]_0^1 = 3x^2, 0 \le x \le 1$$



Marginal distribution function of Y,

$$f_Y(y) = \int_0^1 f_{X,Y}\ (x,y)\, dx = \int_0^1 6x^2y\, dx = \frac{6y}{3}[x^3]_0^1 = 2y,\ 0 \le y \le 1$$

Solution

The marginal cumulative distribution function of X is obtained by integrating the marginal PDF from 0 to x,

$$F_X(x) = \int_0^x f_X(x)\,dx = \int_0^x 3x^2\,dx = \frac{3}{3}[x^3]_0^x = x^3, 0 \le x \le 1$$

Similarly, the marginal cumulative distribution function of Y is obtained as,

$$F_Y(y) = \int_0^y f_Y(y)\,dy = \int_0^y 2y\,dx = \frac{2}{2}[y^2]_0^1 = y^2,\ 0 \le y \le 1$$

Example – 2

A data scientist is tuning a Convolutional Neural Network (CNN). Let the continuous random variables X and Y represent the normalized Dropout Rate and Learning Rate, respectively (ranging from 0 to 1). The joint probability density function (PDF) defining the likelihood of success in this search space is given by:

$$f(x,y) = \begin{cases} C(x+y^2), & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & otherwise \end{cases}$$

(a) Determine the value of the normalization constant C.

(b) Find the marginal distribution function of Dropout Rate (X) and Learning Rate (Y).

Solution

(a) For $f\ (x,y)$ to be a valid joint probability density function,

$\int_0^1 \int_0^1 f\ (x,y)\,dx\,dy = 1$

Given joint pdf is, $f\ (x,y) = C(x+y^2),\ 0 \le x \le 1, 0 \le y \le 1$

$$\Rightarrow \int_0^1 \int_0^1 C(x+y^2)\,dx\,dy = 1$$

$$\Rightarrow \int_0^1 C\left[\frac{x^2}{2} + y^2 x\right]_0^1 dy = 1$$

$$\Rightarrow \int_0^1 C\left[\frac{1}{2} + y^2\right] dy = 1$$

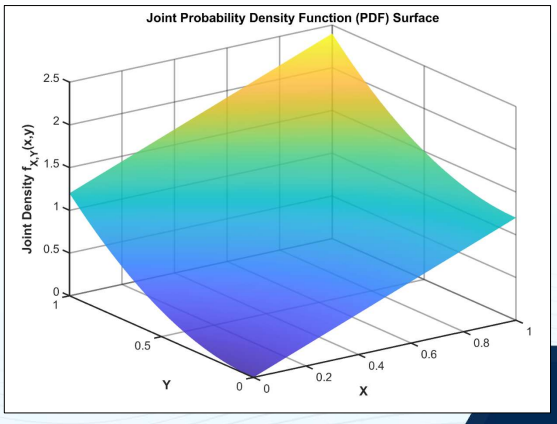
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Solution

$$\Rightarrow C\left[\frac{y}{2} + \frac{y^3}{3}\right]_0^1 = 1$$

$$\Rightarrow C\left[\frac{5}{6}\right] = 1$$

$$\Rightarrow C = \frac{6}{5}$$



- Thus, the joint pdf is,

$$f\ (x,y) = \frac{6}{5}(x+y^2), \qquad 0 \le x \le 1, 0 \le y \le 1$$

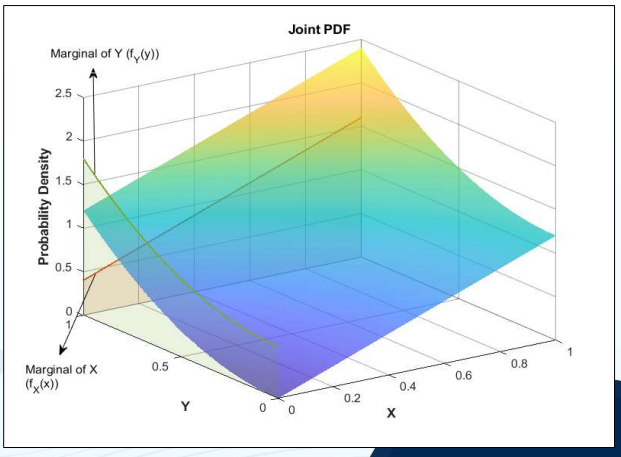
Solution

(b) Marginal distribution function of Dropout Rate (X) is calculated as,

$$f_X(x) = \int_0^1 f_{X,Y}(x,y)\,dy$$

$$= \int_0^1 \frac{6}{5}(x+y^2)\,dy = \frac{6}{5}\left[xy + \frac{y^3}{3}\right]_0^1$$

$$= \frac{6}{5}\left[x + \frac{1}{3}\right]$$



So, Marginal distribution function of Dropout Rate (X) is,

$$f_X(x) = \frac{6}{5}\left[x + \frac{1}{3}\right], 0 \le x \le 1$$

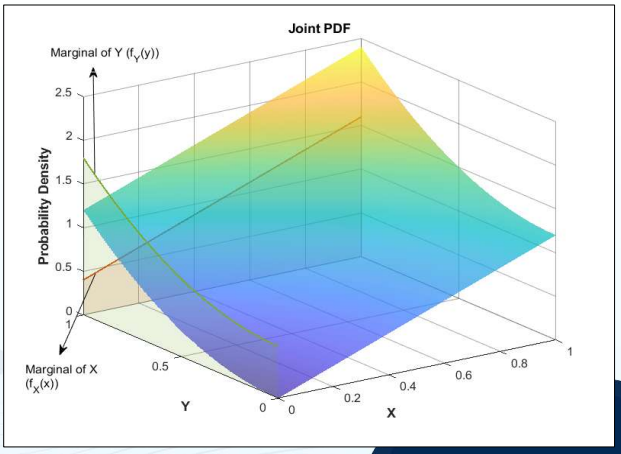
Solution

Marginal distribution function of Learning Rate (Y) is calculated as,

$$f_Y(y) = \int_0^1 f_{X,Y}(x,y)\,dx = f_Y(y)$$

$$= \int_0^1 \frac{6}{5}(x+y^2)\,dx = \frac{6}{5}\left[\frac{x^2}{2} + y^2 x\right]_0^1$$

$$= \frac{6}{5}\left[\frac{1}{2} + y^2\right]$$



Therefore, Marginal distribution function of Learning Rate (Y) is,

$$f_Y(y) = \frac{6}{5}\left[\frac{1}{2} + y^2\right], 0 \le y \le 1$$


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ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 04 : Marginal and Conditional Probability Distributions

LECTURE W4L4 : Conditional Probability Distributions

BY

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CONCEPTS COVERED

Conditional Probability Distributions: Discrete and Continuous Random Variables

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Conditional Probability Distribution

- Conditional probability distribution is one of the important concepts in multivariate probability distributions.
- It may be recalled that marginal distribution of a random variable completely marginalizes out the other variable.
- However, conditional probability distribution of a random variable is the probability distribution of one variable (say X) for a specific value/range of another variable (say Y).
- For example, the distribution of X given $Y = y_0$ and the distribution of Y given $x_1 \leq X \leq x_2$.
- Let us also recall the conditional probability of event A, conditioned on event B is expressed as,

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Conditional Probability Distribution

- This conditional probability theorem can be utilized to obtain the conditional distribution function of the random variables.
- Conditional distribution concept is explained using bivariate distributions involving two random variables.
- The same principles can later be extended to multivariate systems involving several random variables.
- This concept is widely applied in prediction, modelling, and risk assessment studies.

Conditional Probability Distribution

Discrete Random Variable:

- Let X and Y be two discrete random variables and we have the events $(X = x)$ and $(Y = y)$.
- By using the conditional probability theorem, the conditional probability of Y given X under the said conditions can be expressed as,

$$P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P(X = x)}$$

where $P(X = x, Y = y)$ is the joint pmf of X and Y, i.e., $p_{X,Y}(X = x, Y = y) = p_{X,Y}(x, y)$ and $P(X = x)$ is the marginal probability distribution of X i.e., $p_X(X = x) = p_X(x)$.

Discrete Random Variable:

- Thereby, using joint pmf and the marginal probability distribution, the conditional distribution function of Y given X can be expressed as,

$$p_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)} \qquad \text{if } p_X(x) \neq 0$$

- Similarly, the conditional probability of X given $Y = y$ can be expressed as,

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} \qquad \text{if } p_Y(y) \neq 0.$$

Discrete Random Variable:

- The same concept can be further extended by considering the condition over a set or range of values, such as $X \in \tilde{R}$, where $\tilde{R}$ may include multiple values of X.
- In such cases, the conditional distribution function of Y given $X \in \tilde{R}$ can be expressed as:

$$p_{Y/X}(y/X \in \tilde{R}) = \frac{\sum_{x_i \in \tilde{R}} p_{X,Y}(x_i, y)}{\sum_{x_i \in \tilde{R}} p_X(x_i)}$$

- Similarly, the conditional distribution function of X given $Y \in \tilde{R}$ can be expressed as:

$$p_{X/Y}(x/Y \in \tilde{R}) = \frac{\sum_{y_j \in \tilde{R}} p_{X,Y}(x, y_j)}{\sum_{y_j \in \tilde{R}} p_Y(y_j)}$$

Example-1

Severity of a natural extreme event (say, drought) at two study regions is categorized into five different categories, i.e., 1, 2, 3, 4, and 5. Let the random variables X and Y denote these observed categories for the first and second study regions, respectively. The joint pmf is provided in the following table. Estimate the conditional probability of Y given $X \geq 4$.

Random Variable		Y				
		1	2	3	4	5
X	1	0.20	0.05	0.00	0.00	0.00
	2	0.05	0.30	0.05	0.00	0.00
	3	0.03	0.07	0.09	0.04	0.00
	4	0.01	0.01	0.02	0.02	0.01
	5	0.01	0.00	0.01	0.02	0.01

Solution

The conditional probability of Y given $X \geq 4$ is evaluated using the conditional probability theorem as follows:

$$P(Y|X \geq x) = \frac{\sum_{x_i \geq x} P_{X,Y}(x_i, y)}{\sum_{x_i \geq x} P_X(x_i)}$$

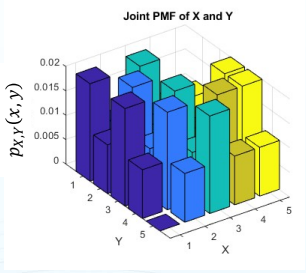
$$\Rightarrow P(Y|X \geq 4) = \frac{\sum_{x_i \geq 4} P_{X,Y}(x_i, y)}{\sum_{x_i \geq 4} P_X(x_i)}$$

The denominator represents the marginal probability corresponding to $X \geq 4$.

Solution

Marginal probability corresponding to $X \geq 4$ is evaluated as,

Random Variable		Y				
		1	2	3	4	5
X	1	0.20	0.05	0.00	0.00	0.00
	2	0.05	0.30	0.05	0.00	0.00
	3	0.03	0.07	0.09	0.04	0.00
	4	0.01	0.01	0.02	0.02	0.01
	5	0.01	0.00	0.01	0.02	0.01



$$\sum_{x_i \geq 4} P_X(x_i) = 0.010 + 0.010 + 0.020 + 0.020 + 0.010$$

$$+0.010 + 0 + 0.010 + 0.020 + 0.010 = 0.12$$

Solution

Thus, $\sum_{x_i \geq 4} P_X(x_i) = 0.12$

The numerator values for each value of y_i are obtained from the joint probability table.

$X \backslash Y$	1	2	3	4	5
1	0.20	0.05	0.00	0.00	0.00
2	0.05	0.30	0.05	0.00	0.00
3	0.03	0.07	0.09	0.04	0.00
4	0.01	0.01	0.02	0.02	0.01
5	0.01	0.00	0.01	0.02	0.01
$\sum_{x_i \geq 4} P_{X,Y}(x_i, y)$	0.01+0.01 =0.02	0.01+0 =0.01	0.02+0.01 =0.03	0.02+0.02 =0.04	0.01+0.01 =0.02

Solution

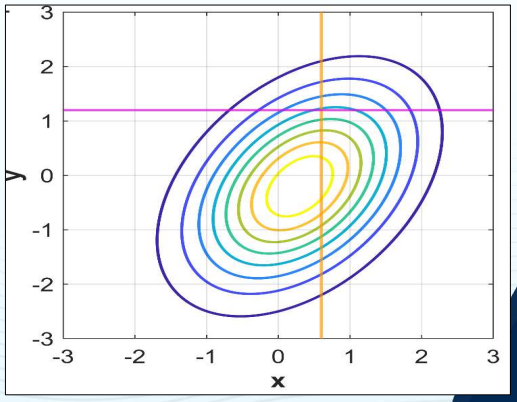
By knowing the denominator and numerator of the conditional probability $P(Y|X \geq 4)$, the probability of Y given $X \geq 4$ can be evaluated for each y_i as follows:

Y	1	2	3	4	5
$\frac{\sum_{x_i \geq 4} P_{X,Y}(x_i,y)}{\sum_{x_i \geq 4} P_X(x_i)}$	$\frac{0.02}{0.12}$	$\frac{0.01}{0.12}$	$\frac{0.03}{0.12}$	$\frac{0.04}{0.12}$	$\frac{0.02}{0.12}$
$P(Y\|X \geq 4)$	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{6}$

Conditional Distribution Function

Continuous Random Variable:

- For continuous random variables, the concept of conditional distribution remains similar to the discrete random variable.
- Let X and Y be two continuous random variables having a joint pdf $f_{X,Y}(x,y)$ with marginal distributions as $f_X(x)$ and $f_Y(y)$.



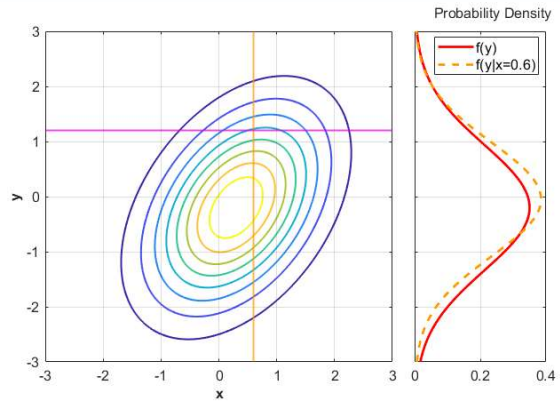
Joint pdf $f_{X,Y}(x,y)$

Continuous Random Variable:

- The conditional probability distributions of Y given $X = x$ is:

$$f_{Y/X}(y/x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

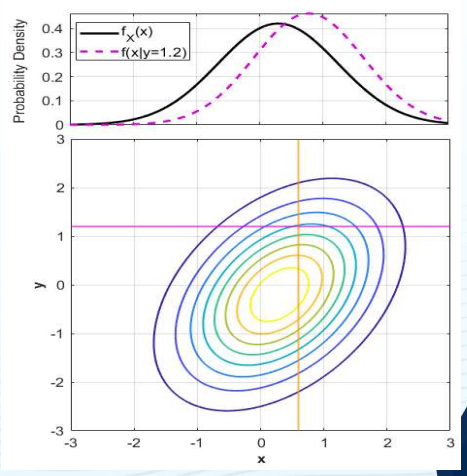
- In the above expression, the conditioning variable remains as an unspecified value (x) in the conditional distribution function.
- This is especially convenient when it is required to derive conditional distributions for several different values of the conditioning variable.



Continuous Random Variable:

- Similarly, the conditional distribution function of X given $Y = y$ can be expressed as:

$$f_{X/Y}(x/y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$



Continuous Random Variable:

- When the conditioning variable belongs to a range, say $\tilde{R} \equiv x_1 \leq X \leq x_2$, the conditional distribution function of Y, conditioned on $X \in R$ can be expressed as:

$$f_{Y/X}(y/x_1 \leq X \leq x_2) = \frac{\int_{x_1}^{x_2} f_{X,Y}(x,y)dx}{\int_{x_1}^{x_2} f_X(x)dx}$$

- Similarly, the expression of the conditional distribution function of X, conditioned on $Y \in \tilde{R}$, where $\tilde{R} \equiv y_1 \leq Y \leq y_2$, is expressed as:

$$f_{X/Y}(x/y_1 \leq Y \leq y_2) = \frac{\int_{y_1}^{y_2} f_{X,Y}(x,y)dy}{\int_{y_1}^{y_2} f_Y(y)dy}$$

Conditional CDF for Continuous RV

- The conditional CDF, $F_{Y/X}(y/x)$ gives the probability that $Y \leq y$, when $X = x$ is known, which can be expressed as:

$$F_{Y/X}(y/x) = \int_{-\infty}^{y} \frac{f_{X,Y}(x,y)}{f_X(x)} dy$$

- Similarly, the conditional CDF, $F_{X/Y}(x/y)$ gives the probability that $X \leq x$, when $Y = y$ is known, which can be expressed as:

$$F_{X/Y}(x/y) = \int_{-\infty}^{x} \frac{f_{X,Y}(x,y)}{f_Y(y)} dx$$

Example-2

Let the random variables X and Y denote the False Positive Rate (FPR) and True Positive Rate (TPR), respectively, for a binary classification algorithm. For this model, the TPR is constrained to exceed the FPR. The joint probability density function of these classification metrics, i.e. FPR (X) and TPR (Y) is defined as:

$$f_{X,Y}(x,y) = \begin{cases} 6x & 0 < x < y < 1 \\ 0 & otherwise \end{cases}$$

Evaluate the conditional distribution of Y given X.

Solution

The joint probability density function of FPR (X) and TPR (Y) is defined as:

$$f_{X,Y}(x,y) = \begin{cases} 6x & 0 < x < y < 1 \\ 0 & otherwise \end{cases}$$

To evaluate the conditional distribution of Y given X, the marginal distribution of X needs to be calculated.

The marginal distribution of X is obtained by integrating the joint PDF over all possible values of Y.

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy = \int_x^1 6xdy = 6x\int_x^1 dy$$
$$\Rightarrow f_X(x) = 6x[y]_x^1 = 6x(1-x)$$

Thus, the marginal distribution of X, $f_X(x) = 6x(1-x)$

Solution

Conditional distribution of Y given X :

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{6x}{6x(1-x)} = \frac{1}{(1-x)}$$

Thus, conditional distribution of TPR (Y) given FPR (X) is,

$$f_{Y|X}(y \mid x) = \begin{cases} \frac{1}{1-x} & x < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Example-3

To evaluate a Neural Network based forecasting model, two normalized scores are analyzed: number of parameters X (which quantifies complexity of the model), and the prediction error score Y (both normalized to the range between 0 and 1). Their joint probability density function (pdf) is defined as:

$$f_{X,Y}(x,y) = \begin{cases} 4\left(xy^2 + \frac{x^2y}{2}\right) & 0 \le x \le 1, 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

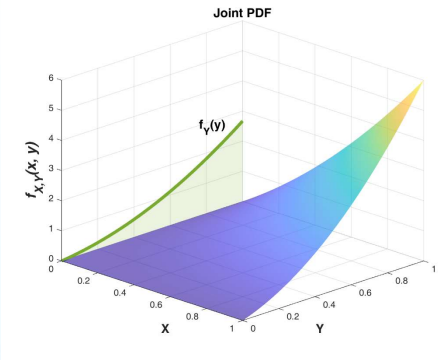
Find the probability that the model complexity score is less than 0.2, given that the prediction error is 0.2, i.e., $P(x < 0.2 | y = 0.2)$.

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Solution

Given joint PDF is:

$$f_{X,Y}(x,y) = \begin{cases} 4\left(xy^2 + \frac{x^2y}{2}\right) & 0 \le x \le 1, 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$



To evaluate the conditional distribution of X given Y, the marginal distribution of Y needs to be calculated.

The marginal distribution of Y is,

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dx = \int_0^1 4\left(xy^2 + \frac{x^2y}{2}\right)dx$$

$$\Rightarrow f_Y(y) = 4\left(\frac{y^2}{2} + \frac{y}{6}\right) = 2y^2 + \frac{2y}{3}, \quad 0 \le y \le 1$$

Solution

Now, the conditional probability of X given Y is expressed as:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

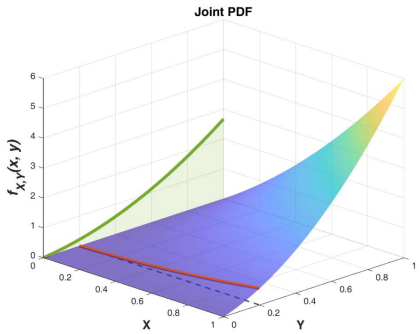
$$\Rightarrow f_{X|Y}\ (x|y = 0.2) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{4\left(xy^2 + \frac{x^2 y}{2}\right)}{4\left(\frac{y^2}{2} + \frac{y}{6}\right)} = \frac{0.2x + \frac{x^2}{2}}{0.1 + \frac{1}{6}}$$

$$\Rightarrow f_{X|Y}\ (x|y = 0.2) = 3.75\left(0.2x + \frac{x^2}{2}\right)$$

$$\Rightarrow f_{X|Y}(x < 0.2|y = 0.2) = \int_0^{0.2} 3.75\left(0.2x + \frac{x^2}{2}\right) dx$$

$$= 3.75\left[\frac{0.2x^2}{2} + \frac{x^3}{6}\right]_0^{0.2} = 0.02$$

Thus, probability that the model complexity score is less than 0.2, given that the prediction error is 0.2 is 0.02.







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ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 04 : Marginal and Conditional Probability Distributions

LECTURE W4L5: Independence between Random Variables

BY

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CONCEPTS COVERED

Independence between Random Variables

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Independence Between Random Variables

- Two random variables, X and Y are said to be independent when the occurrence of one variable does not influence other.
- Mathematically, Two random variables, X and Y are independent if and only if their joint *pmf*, $p_{X,Y}(x,y)$ can be expressed by multiplying the respective marginal probability density finctions as follows:
  - For discrete RVs,
    $$p_{X,Y}(x,y) = p_X(x) \times p_Y(y)$$
  - For continuous RVs,
    $$f_{X,Y}(x,y) = f_X(x) \times f_Y(y)$$

Independence Between Random Variables

- The conditional probability of X and Y becomes equal to their respective marginal distributions, i.e.,
  - For discrete RVs,
    $$p_{Y|X}(y|x) = p_Y(y) \quad \text{and} \quad p_{X|Y}(x|y) = p_X(x)$$
  - For continuous RVs,
    $$f_{X|Y}(x|y) = f_X(x) \quad \text{and} \quad f_{Y|X}(y|x) = f_Y(y)$$

Independence Between Random Variables

- Thus, we may also write:
- In case of discrete RVs

  Thus, $p_{X,Y}(x,y) = p_{X|Y}(x|y) \times p_Y(y) = p_X(x) \times p_{Y|X}(y|x)$

- Similarly, for continuous RVs,

$$f_{X,Y}(x,y) = f_X(x) \times f_Y(y)$$

- Also,

$$f_{X,Y}(x,y) = f_{X|Y}(x|y) \times f_Y(y) = f_X(x) \times f_{Y|X}(y|x)$$

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Example-1

A machine learning model generates predictions using data from two sensors. Let X represent the prediction outcome (Incorrect/Correct) and Y represent the sensor used (Sensor A/Sensor B).The joint probability distribution of X and Y is given below.

$X \backslash Y$	Sensor A (1)	Sensor B (2)
Incorrect (0)	0.12	0.18
Correct (1)	0.28	0.42

Determine whether the prediction outcome (X) and the sensor used (Y) are independent.

Solution

To determine whether X and Y are independent, we need to verify whether

$$p_{X,Y}(x, y) = p_X(x)p_Y(y)$$

for all possible values of x and y.

Marginal probabilities of X are:

$$P(X = 0) = 0.12 + 0.18 = 0.30$$
$$P(X = 1) = 0.28 + 0.42 = 0.70$$

Marginal probabilities of Y are:

$$P(Y = 1) = 0.12 + 0.28 = 0.40$$
$$P(Y = 2) = 0.18 + 0.42 = 0.60$$

Solution

For X and Y to be independent random variables, the independence condition must be satisfied for each possible (x, y) combination.

- $P(X = 0, Y = 1) = 0.12$

  $$P(X = 0) \times P(Y = 1) = 0.3 \times 0.4 = 0.12$$

- $P(X = 0, Y = 2) = 0.18$

  $$P(X = 0) \times P(Y = 2) = 0.3 \times 0.6 = 0.18$$

- $P(X = 1, Y = 1) = 0.28$

  $$P(X = 1) \times P(Y = 1) = 0.7 \times 0.4 = 0.28$$

Solution

- $P(X = 1, Y = 2) = 0.42$

$$P(X = 1) \times P(Y = 2) = 0.7 \times 0.6 = 0.42$$

- Since the joint probabilities of all the combinations is equal to their product of their marginal probabilities, the prediction outcome (X) and the sensor used (Y) are independent.

Example-2

An image recognition model extracts two feature scores from an image. Let X represent the edge-feature score and Y represent the texture-feature score. (both scores range from 0 to 1). The joint probability density function of X and Y is given by,

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2), & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & otherwise \end{cases}$$

Determine whether the two feature scores are independent.

Solution

Given joint PDF:

$$f(x,y) = \begin{cases} \frac{6}{5}(x+y^2), & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & otherwise \end{cases}$$

To determine whether X and Y are independent, we need to verify whether

$$f_{X,Y}(x,y) = f_X(x) \times f_Y(y)$$

Marginal distribution function of X,

$$f_X(x) = \int_0^1 f_{X,Y}(x,y)\,dy = \int_0^1 \frac{6}{5}(x+y^2)\,dy = \frac{6}{5}\left[xy + \frac{y^3}{3}\right]_0^1$$

Solution

$$f_X(x) = \frac{6}{5}\left[x + \frac{1}{3}\right], 0 \le x \le 1$$

Marginal distribution function of Y,

$$f_Y(y) = \int_0^1 f_{X,Y}(x,y)\,dx = \int_0^1 \frac{6}{5}(x+y^2)\,dx = \frac{6}{5}\left[\frac{x^2}{2} + y^2 x\right]_0^1 = \frac{6}{5}\left[\frac{1}{2} + y^2\right], 0 \le y \le 1$$

$$\Rightarrow f_X(x) \times f_Y(y) = \frac{6}{5}\left[x + \frac{1}{3}\right] \times \frac{6}{5}\left[\frac{1}{2} + y^2\right] = \frac{36}{25}\left[x + \frac{1}{3}\right]\left[\frac{1}{2} + y^2\right]$$

Thus, $f_X(x) \times f_Y(y) \neq f_{X,Y}(x,y)$

Hence, edge-feature score and texture-feature score are not independent.

Independence Between Random Variables

- If the RVs are independent, their joint pmf/cdf can be obtained by multiplying their respective marginal distributions.
- However, if the variables are dependent, it is not easy to obtain their joint distribution. In such cases, copula theory may be used to derive their joint distribution.

